Semantic Dissonance Detection: A Contrastive Kinematic Transformer for Predictive Support for Errorless Learning in VR

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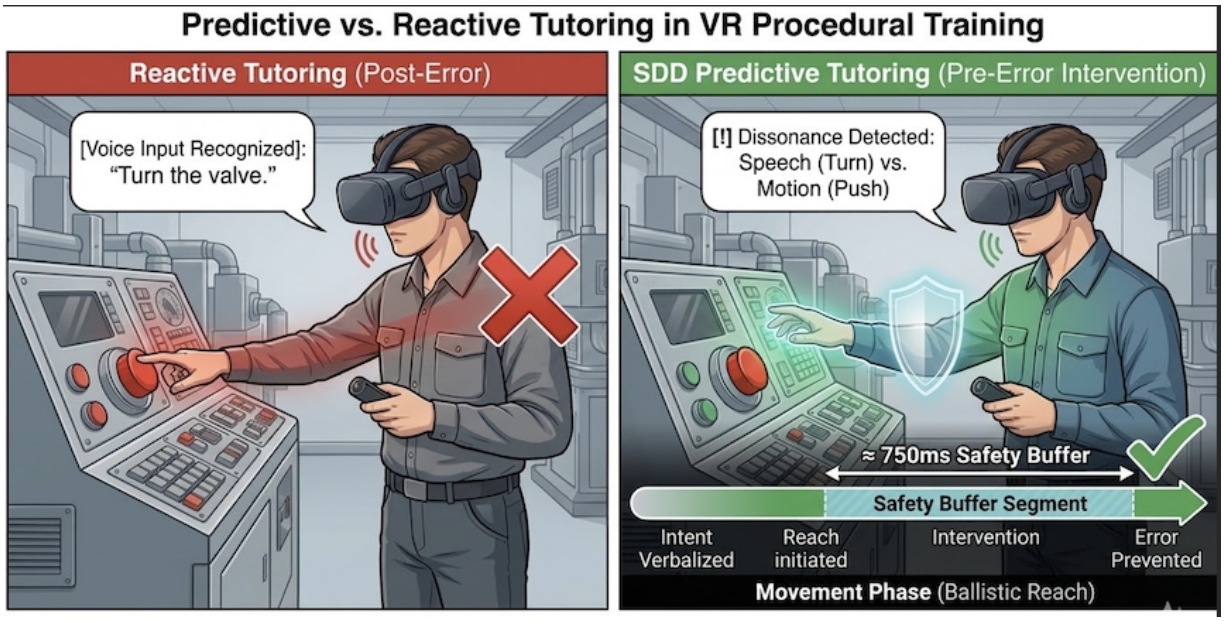


Figure 1: Predictive versus reactive tutoring in VR procedural training. Traditional systems flag errors post-execution (left). The proposed Semantic Dissonance Detection framework evaluates intent-action divergence during the ballistic reach phase, providing an approximate 750ms pre-emptive safety buffer (right).

Abstract

Virtual Reality procedural training currently relies on reactive tutoring, which corrects users only after errors occur, risking negative training transfer. We introduce Semantic Dissonance Detection, a cross-modal framework that predicts and intercepts action slips pre-emptively. By continuously comparing a user's verbalized intent (encoded via a Vision-Language Model) against real-time OpenXR kinematics (processed by a custom Contrastive Transformer), the system estimates intent-action divergence. In a 50-user study, our framework achieved 93.7% Leave-One-Subject-Out accuracy. During applied testing, it provided a 750ms predictive safety buffer, yielding a 95.3% relative reduction in completed physical errors by enabling mid-reach correction.

Index Terms: Virtual Reality, Intelligent Tutoring Systems, Contrastive Learning, Kinematic Tracking, Action Slip Detection, Human-Computer Interaction.

1 Introduction

Procedural training in Virtual Reality (VR) is increasingly adopted for high risk operational domains including aviation, surgical simulation, and industrial maintenance [2]. A primary pedagogical advantage of VR is the ability to safely fail. However, standard Intelligent Tutoring Systems (ITS) deployed in these environments operate reactively. They evaluate discrete event triggers such as physics collisions. If a trainee is instructed to activate an abort button but mistakenly reaches for a slide throttle, the ITS only registers the anomaly after the throttle is physically manipulated.

This reactive loop exposes a critical flaw in spatial motor skill acquisition. Allowing a user to complete an erroneous physical motion builds negative muscle memory, a phenomenon known in human factors engineering as negative training transfer [1]. In high reliability organizations, training must not only teach the correct procedure but actively prevent the encoding of incorrect motor routines. To achieve the errorless learning paradigm, where mistakes are intercepted before they are fully actualized, the system must predict the error during the early ballistic reach phase [9].

We propose Semantic Dissonance Detection (SDD). SDD is a predictive, cross-modal framework that continuously compares what the user intends to do against what their body is physically doing in three-dimensional space. We extract intent via speech recognition and encode it using the pre-trained CLIP Vision-Language Model [10]. Concurrently, we feed the raw OpenXR controller data into a custom Kinematic Transformer trained via Supervised Contrastive Learning (SupCon) [5]. This mathematically forces the continuous representations of physical human motion to align with the discrete semantic text embeddings of CLIP.

The primary contributions of this research are:

- A cross-modal architecture mapping raw VR kinematic data to a semantic text space, reducing reliance on explicit object-

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level collision tracking.

- The formulation of Relative Position Normalization for sliding window sequence modeling to ensure local translational invariance.
- An empirical evaluation (N=50) demonstrating a 93.7% Leave-One-Subject-Out accuracy, heavily outperforming modern sequence models and spatial heuristics.
- An applied pedagogical study demonstrating that the framework's 750ms predictive safety buffer was associated with a 95.3% relative reduction in completed physical action slips.

2 Theoretical Foundations

2.1 Cognitive Models of Human Error

In human factors psychology, James Reason's Generic Error-Modeling System divides human fallibility into mistakes and slips [11]. Mistakes are failures of planning. Slips are failures of execution [7]. In VR training, novices frequently commit action slips. They possess the correct semantic intent but output the wrong physical trajectory. Because the intent is correct, post-hoc reactive corrections are highly frustrating. The pedagogical system must therefore intercept the execution failure before the physical loop closes.

2.2 Motor Control and Trajectory Prediction

The feasibility of predicting an action slip within the first 250 milliseconds relies heavily on the biological mechanics of human motor control. According to the Minimum Jerk Model [4], a goal-directed ballistic reach consists of an initial high-acceleration impulse followed by a deceleration phase guided by visual feedback. The initial trajectory vector contains the highest signal-to-noise ratio regarding the user's intended spatial target. By capturing the initial acceleration phase, the Kinematic Transformer can decode the intended terminal constraint without waiting for the user to complete the deceleration phase mandated by Fitts's Law [3].

3 The SDD Architecture

The SDD framework operates as a dual-stream pipeline converging at a temporal dissonance metric, as detailed in Figure 2.

3.1 Semantic Stream (Intent Encoding)

VR procedural actions are better represented as continuous vectors in a multimodal semantic space rather than discrete categorical outputs. The user's speech is transcribed to text via standard OS-level APIs. We utilize the frozen text encoder of the CLIP model to generate the semantic anchor [10]. The text string is projected into a dense 512-dimensional embedding space. Using CLIP enables future extension to open-vocabulary procedural actions without retraining semantic representations. Because the intent is declared strictly prior to movement, this vector $\mathbf{z}_{text} \in \mathbb{R}^{512}$ remains static throughout the reach execution. This capability assumes that virtual objects within the scene are inherently tagged with descriptive semantic metadata, a standard practice in game engine development.

3.2 Kinematic Stream (Action Encoding)

The physical action is captured continuously via the OpenXR runtime. At any discrete time t, the spatial state vector $\mathbf{s}_t$ is defined as a 13-element array encompassing global position $\mathbf{p}$, linear velocity $\mathbf{v}$, angular velocity ω, and quaternion rotation $\mathbf{q}$. Quaternions are utilized exclusively over Euler angles to avoid gimbal lock singularities [12].

To capture the temporal dynamics of the human motor system, we maintain a continuous rolling temporal window W of length $N = 45$ frames. At a standard 90Hz polling rate, this encapsulates exactly 0.5 seconds of dense movement data. Evaluating a reach at the 250-millisecond mark simply means the window contains roughly 250 milliseconds of pre-motor resting state data and 250 milliseconds of active ballistic acceleration.

3.2.1 Relative Position Normalization

To enforce strict translational invariance, we apply Relative Position Normalization at the initiation of every frame window. We subtract the position vector of the initial frame $\mathbf{p}_0$ from all subsequent positional frames $\mathbf{p}_i$ within that specific window. Linear and angular velocities remain unaltered.

3.3 Contrastive Kinematic Transformer

The raw 13-dimensional input is linearly projected to a 128-dimensional representation. We inject a fixed Sinusoidal Positional Encoding to retain the temporal ordering of the motor execution [13]. The core processing is handled by three layers of Multi-Head Self-Attention. Crucially, while the Relative Position Normalization naturally zeroes out the spatial coordinates during the initial pre-motor resting phase of a reach, the self-attention formulation assigns low attention weights to these static frames and shifts its focus entirely to the active ballistic gradient. Average pooling collapses the temporal data into a vector, which is mapped to match the CLIP semantic space, producing $\mathbf{z}_{kin}(t) \in \mathbb{R}^{512}$.

3.4 Supervised Contrastive Alignment (SupCon)

Standard InfoNCE contrastive learning [8] artificially caps spatial accuracy by penalizing identical action classes in randomized batches. We utilize Supervised Contrastive Loss [5] to encourage clustering of identical physical motions. We define I as the set of all indices in a multi-modal batch, and $P(i)$ as the set of positive matches distinct from i. The cross-modal SupCon loss mapping kinematics toward text is formulated as:

$$\mathscr{L}_{k\to t} = \sum_{i\in I} \frac{-1}{|P(i)|} \sum_{p\in P(i)} \log \frac{\exp(\mathbf{z}^i_{kin} \cdot \mathbf{z}^p_{text}/\tau)}{\sum_{a\in I} \exp(\mathbf{z}^i_{kin} \cdot \mathbf{z}^a_{text}/\tau)} \qquad (1)$$

where $\tau = 0.07$. The symmetric loss $\mathscr{L}_{t\to k}$ is calculated identically. The final total loss applied during backpropagation is their mean $\mathscr{L}_{total} = (\mathscr{L}_{k\to t} + \mathscr{L}_{t\to k})/2$.

4 Methodology and User Study

4.1 Apparatus and Experimental Protocol

To validate the framework, we conducted a data collection study involving N=50 participants (Meta Quest 3, 90Hz polling). To rigorously test the bounds of the SDD architecture, we constrained our experimental protocol to four highly confusable spatial primitives. Participants stood in front of a virtual control panel containing a Push button (Linear Z-axis), a Pull lever (Inverse Linear Z-axis), a Turn valve (Angular local-axis), and a Slide throttle (Linear X-axis).

This protocol forces the network to solve a much harder problem. Push and Slide actions share near-identical biological acceleration curves during the first 250 milliseconds, diverging only by subtle micro-variations in the user's wrist quaternion and spatial approach angle.

4.2 Phase 1 and Phase 2 Execution

During Phase 1, participants verbalized their intended action and subsequently executed it, yielding exactly 1,500 balanced reaches. In Phase 2, an applied learning trial, participants were subjected to cognitive load distractors (specifically, a secondary continuous auditory n-back task) to reliably induce genuine action slips. Participants were randomly assigned to Group A (n=25) or Group B (n=25) in a between-subject design. They utilized either a standard reactive ITS (Group A) or our predictive SDD framework (Group

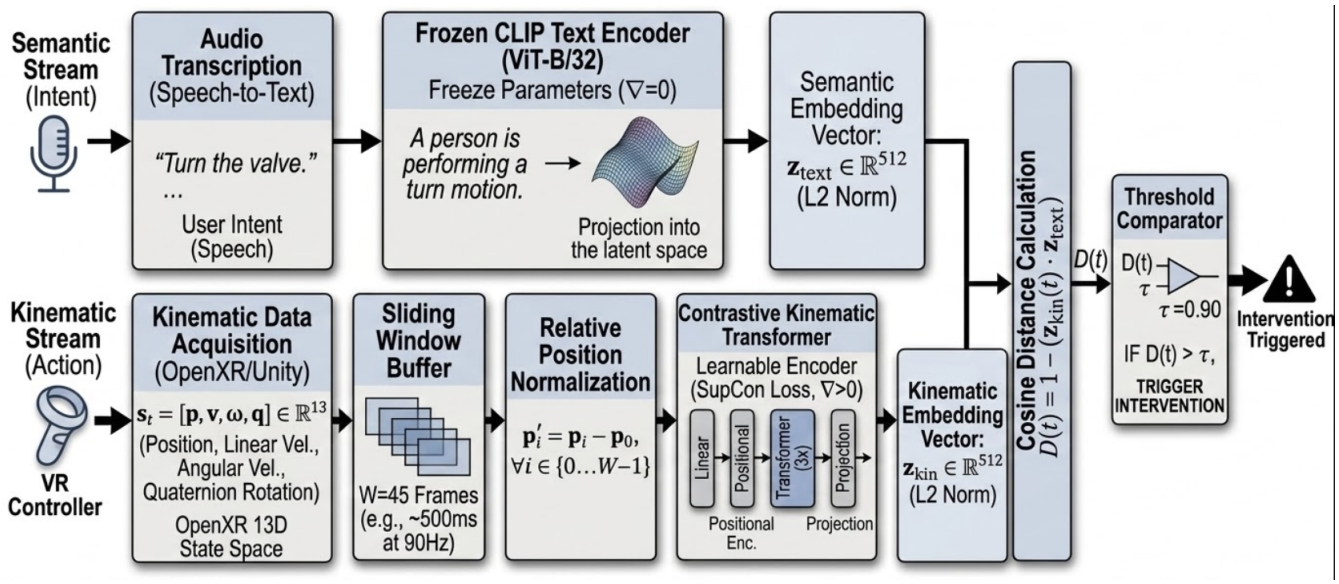


Figure 2: The Semantic Dissonance Detection architecture. Continuous 13-element OpenXR kinematic tracking data is normalized and processed via a Contrastive Transformer. This output is mathematically aligned with the latent semantic space of a frozen Vision-Language Model. The cosine distance between the vectors serves as the real-time dissonance tracking metric.

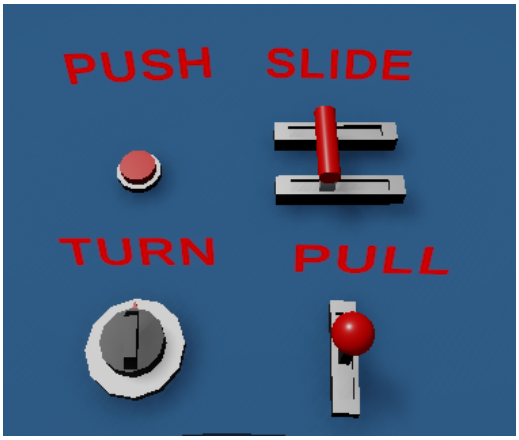


Figure 3: The experimental apparatus. The system evaluates closely related linear and angular movements to test classification bounds.

B). If $D(t) > 0.90$, a visual shield appeared over the objects and maximum-intensity haptic feedback was triggered on the controller.

5 Results

5.1 Classification Performance versus Baselines

To objectively evaluate the representational power of the cross-modal latent space, we compared the offline classification accuracy of the model on the holdout test set. For a fair comparison, all baseline models were strictly evaluated using the same Leave-One-Subject-Out (LOSO) cross-validation protocol.

As detailed in Table 1, utilizing a strict Leave-One-Subject-Out (LOSO) cross-validation, the model achieved 93.7% accuracy. This provides a participant-independent evaluation of cross-user generalization without session-specific overfitting.

We also added a geometric baseline predicting class purely from the early reach direction. The spatial heuristic achieved 54.0% at 150ms and 71.4% at 250ms. In contrast, the Contrastive Transformer achieved 82.5% at 150ms and 94.1% at 250ms. The 94.1% result refers to early prediction at the 250ms observation point, whereas Table 1 reports terminal sequence LOSO accuracy. This confirms that biological acceleration curves and wrist quaternion shifts are required to maintain high accuracy through the ballistic reach.

Table 1: Action Classification Performance on Test Set

Model Architecture	Terminal Accuracy (LOSO)
Spatial Heuristic (Approach Direction Only)	0.642
LSTM (2-Layer)	0.765
Temporal CNN (TCN)	0.841
Standard Transformer (Cross-Entropy)	0.852
Proposed Kinematic Transformer (Contrastive)	**0.937**

5.2 Temporal Dissonance and the Safety Buffer

We evaluated the continuous dissonance trajectory $D(t)$ by comparing congruent trials against simulated incongruent action slips. Incongruent trials were generated by pairing kinematic trajectories with mismatched intent labels.

As illustrated in Figure 5, the alignment converges to a stable congruence baseline of approximately 0.75. The SDD framework consistently triggered the haptic and visual intervention at approximately 250ms. Crucially, at $\tau = 0.90$, the system maintained a low false-positive rate (1.2%) and high precision (98.4%), ensuring that correct reaches were not erroneously interrupted. Against a mean physical reach duration of 1050ms, the system fires approximately 750ms before terminal contact. This establishes an approximately 750ms predictive safety buffer for the user to perceive the intervention and successfully inhibit their motor execution before physical contact occurs.

5.3 Pedagogical Efficacy and Latency Profiling

During Phase 2, participants in Group A completed their erroneous physical reach on 88.5% of all induced action slips. The SDD predictive framework drastically altered user behavior. The 750ms

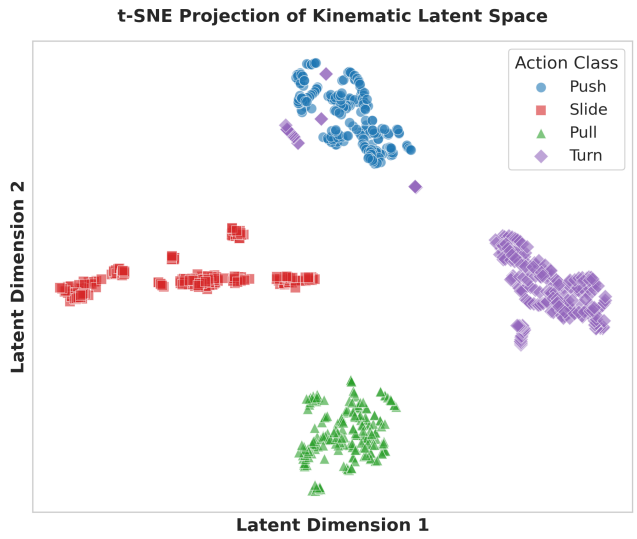


Figure 4: A dense 2D t-SNE projection of the cross-modal latent space using the holdout test data (axes not interpretable, for qualitative visualization). SupCon forces distinct spatial clusters.

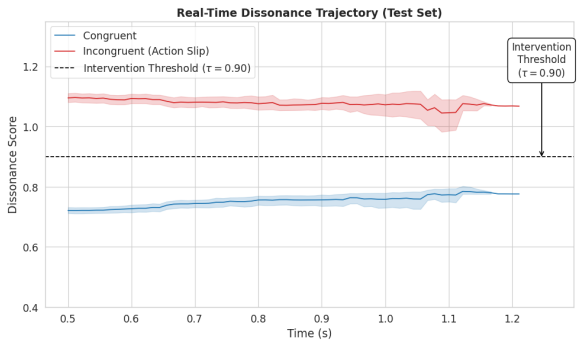


Figure 5: Real-time temporal dissonance trajectory $D(t)$. Incongruent action slips cleanly cross the empirical intervention threshold ($\tau = 0.90$) within the first 0.25 seconds of the physical reach.

safety buffer allowed Group B users to consciously halt and redirect their arms mid-air. Only 4.2% of action slips in Group B resulted in a completed physical error. This was associated with a 95.3% relative reduction in erroneous motor schema consolidation (independent t-test, $p < 0.001$, Cohen's $d = 2.85$).

Operating on a consumer-grade NVIDIA RTX 4080, the sliding window extraction, normalization, and Transformer inference required an average of 8.4 milliseconds per frame, ensuring no cybersickness-inducing frame drops within the 11.1ms VR rendering budget [6].

6 Discussion and Limitations

The high efficacy of the contrastive alignment stems from its ability to separate kinematically similar manifolds. However, a critical limitation of our current evaluation is the fixed virtual UI layout. Because the four interaction targets remained in static positions, the model may partly rely on spatial direction and target-specific kinematic patterns. Demonstrating true semantic generalization independent of spatial priors requires future experiments involving fully randomized UI coordinate layouts.

Furthermore, our reported dissonance trajectories (Figure 5) were thresholded using synthetic incongruent intent-motion pairs. Future work must validate this predictive safety buffer directly on the naturally occurring action slips recorded during Phase 2. Finally, because our applied pedagogical test compared the AI framework (early haptic warning) against a standard reactive ITS (late collision warning), it is difficult to completely isolate the AI model's predictive contribution from the general psychological effect of receiving an early alert. Future studies will include a baseline control group receiving randomized early warnings to better quantify the exact pedagogical value of the semantic prediction.

7 Conclusion

We introduced Semantic Dissonance Detection, a predictive cross-modal framework for Virtual Reality tutoring systems. Evaluated on a large dataset of 1,500 trajectories, the framework achieved a 93.7% LOSO accuracy, outperforming both traditional deep sequence baselines and geometric heuristics. Most importantly, the system's 750ms safety buffer demonstrated the potential to yield a 95.3% relative reduction in completed physical action slips in human trainees. This enables early intervention, shifting VR procedural training away from reactive error correction and towards highly effective predictive error prevention.

References

[1] R. Champney and K. M. Stanney. Training transfer in virtual environments: A framework for analysis. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, 48(22):2713–2717, 2004. doi: 10.1177/154193120404802213

[2] D. Checa and A. Bustillo. A review of immersive virtual reality serious games to enhance learning and training. *Multimedia Tools and Applications*, 79(9):5501–5527, 2020. doi: 10.1007/s11042-019-08348-9

[3] P. M. Fitts. The information capacity of the human motor system in controlling the amplitude of movement. *Journal of Experimental Psychology*, 47(6):381–391, 1954. doi: 10.1037/h0055392

[4] T. Flash and N. Hogan. The coordination of arm movements: An experimentally confirmed mathematical model. *Journal of Neuroscience*, 5(7):1688–1703, 1985. doi: 10.1523/JNEUROSCI.05-07-01688.1985

[5] P. Khosla, P. Teterwak, C. Wang, A. Sarna, Y. Tian, P. Isola, A. Maschinot, C. Liu, and D. Krishnan. Supervised contrastive learning. *Advances in Neural Information Processing Systems*, 33:18661–18673, 2020. doi: 10.48550/arXiv.2004.11362

[6] S. M. LaValle, A. Yershova, M. Katsev, and M. Antonov. Head tracking for the Oculus Rift. In *2014 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 187–194. IEEE, 2014. doi: 10.1109/ICRA.2014.6906584

[7] D. A. Norman. Categorization of action slips. *Psychological Review*, 88(1):1–15, 1981. doi: 10.1037/0033-295X.88.1.1

[8] A. v. d. Oord, Y. Li, and O. Vinyals. Representation learning with contrastive predictive coding. *arXiv preprint arXiv:1807.03748*, 2018. doi: 10.48550/arXiv.1807.03748

[9] P. Prather et al. Errorless learning in rehabilitation. *NeuroRehabilitation*, 16(2):89–94, 2001.

[10] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, et al. Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning*, pp. 8748–8763. PMLR, 2021. doi: 10.48550/arXiv.2103.00020

[11] J. Reason. *Human error*. Cambridge University Press, 1990. doi: 10.1017/CBO9781139062367

[12] K. Shoemake. Animating rotation with quaternion curves. *ACM SIGGRAPH Computer Graphics*, 19(3):245–254, 1985. doi: 10.1145/325165.325242

[13] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, vol. 30, 2017. doi: 10.48550/arXiv.1706.03762